Guided Problem Solving – 5: Solution Key

Foundations of Fluid Mechanics, AM2530

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Problem 1 – Corner flow $\vec{V} = Ax\hat{i} - Ay\hat{j}$, $A = 0.3$ s^{-1}

Particle paths. From $\dot{x} = Ax$ and $\dot{y} = -Ay$:

$$x = x_0 e^{At}, \qquad y = y_0 e^{-At}.$$

Point $a(1,1)$ reaches $x = 3/2$ when $e^{A\tau} = 3/2$:

$$\tau = \frac{\ln 1.5}{0.3} = 1.35 \text{ s}.$$

Stretch factor in x: 3/2; compression factor in y: 2/3.

New vertices ($x' = \frac{3}{2}x_0,\ y' = \frac{2}{3}y_0$):

$$a'\left(\tfrac{3}{2}, \tfrac{2}{3}\right), \quad b'\left(\tfrac{3}{2}, \tfrac{4}{3}\right), \quad c'\left(3, \tfrac{4}{3}\right), \quad d'\left(3, \tfrac{2}{3}\right) \text{ m}.$$

Linear deformation rates.

$$\dot{\varepsilon}_{xx} = \frac{\partial u}{\partial x} = +A = +0.3 \text{ s}^{-1}, \qquad \dot{\varepsilon}_{yy} = \frac{\partial v}{\partial y} = -A = -0.3 \text{ s}^{-1}, \qquad \nabla\cdot\vec{V} = 0.$$

Areas. $A_{abcd} = 1 \times 1 = 1$ m^2, $\quad A_{a'b'c'd'} = \frac{3}{2} \times \frac{2}{3} = 1$ m^2.

Answer

$\tau = 1.35$ s; vertices as above; $\dot{\varepsilon}_{xx} = +0.3$, $\dot{\varepsilon}_{yy} = -0.3$ s^{-1}. Area is unchanged (1 m^2): the flow is incompressible ($\nabla\cdot\vec{V} = 0$), so fluid areas are preserved even though the shape is stretched in x and squeezed in y.

Problem 2 – Simple shear, $h = 4$ mm, $U = 4$ mm/s

Velocity field. $u = \dfrac{U}{h}y = 1.0\,y$ (mm/s, y in mm), $v = 0$.

Positions at $t = 1.5$ s. Displacement $\Delta x = u\,\Delta t = 1.5\,y$ mm. Initial points: $a(1,2)$, $c(3,2)$, $b(2,3)$, $d(2,1)$ mm.

Point	y (mm)	Δx (mm)	New position (mm)
a	2	3.0	$a'(4,2)$
c	2	3.0	$c'(6,2)$
b	3	4.5	$b'(6.5,3)$
d	1	1.5	$d'(3.5,1)$

Line $a'c'$ stays horizontal (length 2 mm). Line $b'd'$ tilts downstream: horizontal offset 3.0 mm over a vertical distance 2.0 mm, i.e. $\arctan(3/2) = 56.3°$ from the vertical.

Angular deformation rate.

$$\dot{\gamma}_{xy} = \frac{\partial u}{\partial y} + \frac{\partial v}{\partial x} = \frac{U}{h} = 1.0 \text{ s}^{-1}.$$

Rotation rate.

$$\omega_z = \frac{1}{2}\left(\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y}\right) = -\frac{1}{2}\frac{U}{h} = -0.5 \text{ rad/s} \quad \text{(clockwise)}, \qquad \zeta_z = 2\omega_z = -1.0 \text{ s}^{-1}.$$

Paddle wheel. It rotates clockwise at 0.5 rad/s. The streamlines are straight, but the fluid on the upper side of the wheel moves faster than on the lower side ($\partial u/\partial y > 0$), which twists the wheel. Rotation depends on the velocity gradient (curl), not on streamline curvature.

Answer

$a'(4, 2)$, $c'(6, 2)$, $b'(6.5, 3)$, $d'(3.5, 1)$ mm; $\dot{\gamma}_{xy} = 1.0$ s^{-1}; $\omega_z = -0.5$ rad/s (clockwise); the paddle wheel rotates clockwise at 0.5 rad/s.

Problem 3 – Channel velocity profiles, $y \in [-h, h]$, $v_c = 3$ m/s

Mean velocity: $\bar{V} = \dfrac{1}{2h}\displaystyle\int_{-h}^{h} u\, dy.$

Profile	Shape	$\bar{V}/v_c$	$\bar{V}$ (m/s)
(a)	parabolic, $u = v_c[1 - (y/h)^2]$	2/3	2.00
(b)	triangular, $u = v_c(1 - \lvert y\rvert/h)$	1/2	1.50
(c)	semi-ellipse, $u = v_c\sqrt{1 - (y/h)^2}$	$\pi/4$	2.36
(d)	core $2h/3$ + linear layers $h/3$	1/2	1.50
(e)	core $2h/3$ + parabolic layers $h/3$	5/9	1.67

Working: (a) $\int_{-h}^{h} v_c(1 - y^2/h^2)dy = \frac{4}{3}v_c h$. (b) triangle area $= \frac{1}{2}(2h)v_c = v_c h$. (c) half-ellipse area $= \frac{\pi}{2}v_c h$. (d) $v_c\frac{2h}{3} + 2\left(\frac{1}{2}v_c\frac{h}{3}\right) = v_c h$. (e) $v_c\frac{2h}{3} + 2\left(\frac{2}{3}v_c\frac{h}{3}\right) = \frac{10}{9}v_c h$.

Shear strain rate and vorticity (for $\vec{V} = u(y)\hat{i}$: $\dot{\gamma}_{xy} = du/dy$, $\zeta_z = -du/dy$).

Profile (a):

$$\frac{du}{dy} = -\frac{2v_c y}{h^2}, \qquad \dot{\gamma}_{xy} = -\frac{2v_c y}{h^2}, \qquad \zeta_z = +\frac{2v_c y}{h^2}.$$

Both vary linearly with y; they are zero at the centreline and largest at the walls, $|du/dy|_{max} = 2v_c/h$.

Profile (d):

Region	du/dy	$\dot{\gamma}_{xy}$	ζ_z
Core $\lvert y\rvert < h/3$	0	0	0
Upper layer $h/3 < y < h$	$-3v_c/h$	$-3v_c/h$	$+3v_c/h$
Lower layer $-h < y < -h/3$	$+3v_c/h$	$+3v_c/h$	$-3v_c/h$

Answer

$\bar{V}_a = 2.0$, $\bar{V}_b = 1.5$, $\bar{V}_c = 2.36$, $\bar{V}_d = 1.5$, $\bar{V}_e = 1.67$ m/s. Profile (a): maximum deformation and vorticity at the walls ($2v_c/h$), none at the centreline. Profile (d): uniform maximum deformation $3v_c/h$ throughout the two boundary layers ($|y| \geq h/3$), and a rigid, non-deforming, irrotational core ($|y| < h/3$).

Problem 4 – Plane Poiseuille flow, $u = U\left[1 - (y/b)^2\right]$

Linear deformation rates. With $v = w = 0$ and $u = u(y)$:

$$\dot{\varepsilon}_{xx} = \frac{\partial u}{\partial x} = 0, \quad \dot{\varepsilon}_{yy} = \frac{\partial v}{\partial y} = 0, \quad \dot{\varepsilon}_{zz} = \frac{\partial w}{\partial z} = 0.$$

Angular deformation rate.

$$\dot{\gamma}_{xy} = \frac{\partial u}{\partial y} + \frac{\partial v}{\partial x} = -\frac{2Uy}{b^2}.$$

Vorticity.

$$\vec{\zeta} = \left(\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y}\right)\hat{k} = \frac{2Uy}{b^2}\,\hat{k}.$$

It is zero at $y = 0$ and has maximum magnitude at the walls $y = \pm b$: $|\vec{\zeta}|_{max} = 2U/b$.

Wall shear. From Newton's law of viscosity at the upper wall,

$$\tau_w = \mu \frac{du}{dy}\Big|_{y=b} = -\frac{2\mu U}{b}, \qquad \zeta_z(b) = \frac{2U}{b} \implies \tau_w = -\mu\,\zeta_z(b).$$

Answer

$\dot{\varepsilon}_{xx} = \dot{\varepsilon}_{yy} = \dot{\varepsilon}_{zz} = 0$; $\dot{\gamma}_{xy} = -2Uy/b^2$; $\vec{\zeta} = (2Uy/b^2)\hat{k}$, maximum $2U/b$ at the walls; $\tau_w = -\mu\zeta_z(b)$.

Problem 5 – Couette–Poiseuille flow with zero net flow

Given

$$u(y) = \frac{V_0}{2}\left(1 + \frac{y}{h}\right) + \frac{1}{2\mu}\left(-\frac{dp}{dx}\right)(h^2 - y^2).$$

Flow rate per unit width.

$$q = \int_{-h}^{h} u\,dy = V_0 h + \frac{1}{2\mu}\left(-\frac{dp}{dx}\right)\frac{4h^3}{3} = V_0 h + \frac{2h^3}{3\mu}\left(-\frac{dp}{dx}\right).$$

Zero net flow ($q = 0$).

$$\left(-\frac{dp}{dx}\right) = -\frac{3\mu V_0}{2h^2} \quad \implies \quad \frac{dp}{dx} = +\frac{3\mu V_0}{2h^2}$$

(an adverse pressure gradient).

Velocity profile for $q = 0$.

$$u(y) = \frac{V_0}{2}\left(1 + \frac{y}{h}\right) - \frac{3V_0}{4h^2}(h^2 - y^2) = \frac{V_0}{4}\left[3\frac{y^2}{h^2} + 2\frac{y}{h} - 1\right].$$

Rotation rate.

$$\frac{du}{dy} = \frac{V_0}{2h}\left(1 + \frac{3y}{h}\right), \qquad \Omega_z = \frac{1}{2}\left(\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y}\right) = -\frac{V_0}{4h}\left(1 + \frac{3y}{h}\right).$$

Zero deformation. $\dot{\gamma}_{xy} = du/dy = 0 \Rightarrow 1 + 3y/h = 0 \Rightarrow y = -h/3$. At this location $u = -V_0/3$ (minimum velocity), and $\Omega_z = 0$ as well.

Answer

$\dfrac{dp}{dx} = \dfrac{3\mu V_0}{2h^2}$; $\quad \Omega_z = -\dfrac{V_0}{4h}\left(1 + \dfrac{3y}{h}\right)$; $\quad$ zero deformation at $y = -h/3$.

Problem 6 – Mushroom deflector on a pipe

Exit speed. $Q = 3$ m^3/s. The stream leaves the annular opening (inner radius $r_1 = 1.8$ m, outer radius $R_2 = 2.0$ m) at 45° below the horizontal. The horizontal projected area is

$$A_{horiz} = \pi(R_2^2 - r_1^2) = \pi(4.0 - 3.24) = 2.388 \text{ m}^2,$$

so the area normal to the stream is

$$A_2 = A_{horiz} \sin 45^\circ = 1.688 \text{ m}^2.$$

(Check: $2\pi \bar{r}\, w_\perp = 2\pi(1.9)(0.2 \sin 45^\circ) = 1.688$ m^2.)

$$V = \frac{Q}{A_2} = \frac{3.0}{1.688} = 1.78 \text{ m/s}.$$

Differential continuity. For steady, axisymmetric, incompressible flow with no swirl,

$$\frac{1}{r}\frac{\partial (rV_r)}{\partial r} + \frac{\partial V_z}{\partial z} = 0.$$

In the radially spreading sheet, $V_r = f(z)/r$, so $rV_r = f(z)$ and $\partial(rV_r)/\partial r = 0$. Hence

$$\frac{\partial V_z}{\partial z} = 0.$$

In-plane area preservation.

$$\dot{\varepsilon}_{rr} = \frac{\partial V_r}{\partial r} = -\frac{f}{r^2} = -\frac{V_r}{r}, \qquad \dot{\varepsilon}_{\theta\theta} = \frac{V_r}{r}, \qquad \dot{\varepsilon}_{rr} + \dot{\varepsilon}_{\theta\theta} = 0.$$

The sheet stretches around the circumference at exactly the rate it is compressed radially, so the area of a marked fluid element in the sheet does not change.

Answer

$A_2 = 1.69$ m^2, $\quad V = 1.78$ m/s; $\quad \partial V_z/\partial z = 0$; $\quad \dot{\varepsilon}_{rr} = -V_r/r$, $\dot{\varepsilon}_{\theta\theta} = +V_r/r$, sum $= 0$.

Problem 7 – Boundary layer $u = Uy/\delta(x)$, $\delta = c\sqrt{x}$

Step 1: $\partial u/\partial x$. With $\dfrac{d\delta}{dx} = \dfrac{c}{2\sqrt{x}} = \dfrac{\delta}{2x}$:

$$\frac{\partial u}{\partial x} = -\frac{Uy}{\delta^2}\frac{d\delta}{dx} = -\frac{Uy}{2x\delta}.$$

Step 2: Continuity.

$$\frac{\partial v}{\partial y} = -\frac{\partial u}{\partial x} = \frac{Uy}{2x\delta}.$$

Step 3: Integrate from the wall with $v(x,0) = 0$:

$$v(x,y) = \int_0^y \frac{U\eta}{2x\delta}\, d\eta = \frac{Uy^2}{4x\delta} \quad \Longrightarrow \quad \frac{v}{U} = \frac{1}{4}\frac{\delta}{x}\left(\frac{y}{\delta}\right)^2.$$

Step 4: Numbers ($x = 0.5$ m, $\delta = 5$ mm, so $\delta/x = 0.01$): $v/U = 0.0025\,(y/\delta)^2$.

y	y/δ	v/U
0	0	0
$\delta/2$	0.5	6.25×10^{-4}
δ	1	2.5×10^{-3}

Why $v \neq 0$. As the boundary layer thickens, the fluid decelerates along x ($\partial u/\partial x < 0$). An incompressible fluid cannot accumulate, so continuity requires $\partial v/\partial y > 0$: the fluid must move away from the plate. This outward flow displaces the streamlines of the outer flow (the displacement effect of the boundary layer).

Answer

$v(x,y) = \dfrac{Uy^2}{4x\delta}$; $v/U = 0$ at the wall, 6.25×10^{-4} at $y = \delta/2$, 2.5×10^{-3} at $y = \delta$.